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OFDM Project – Wireless Receivers

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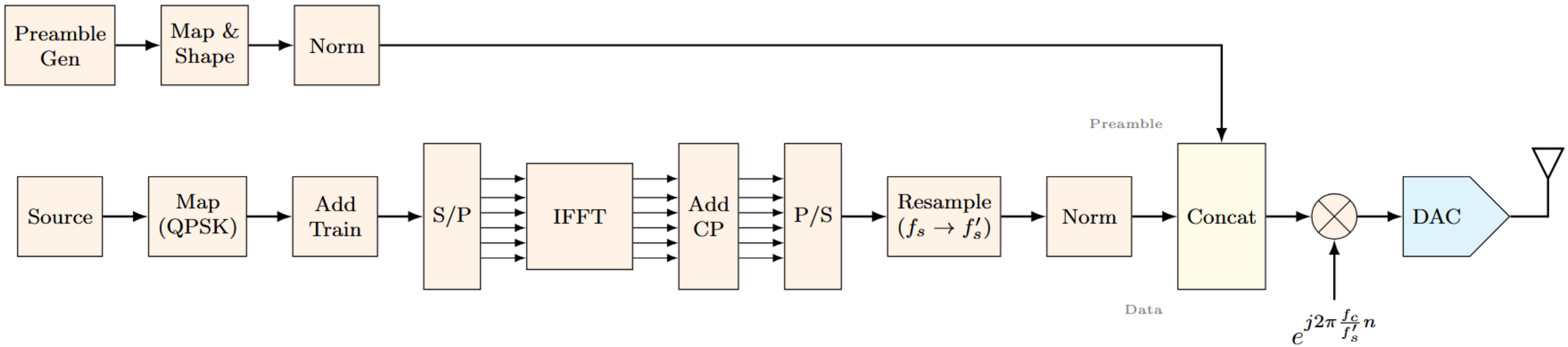
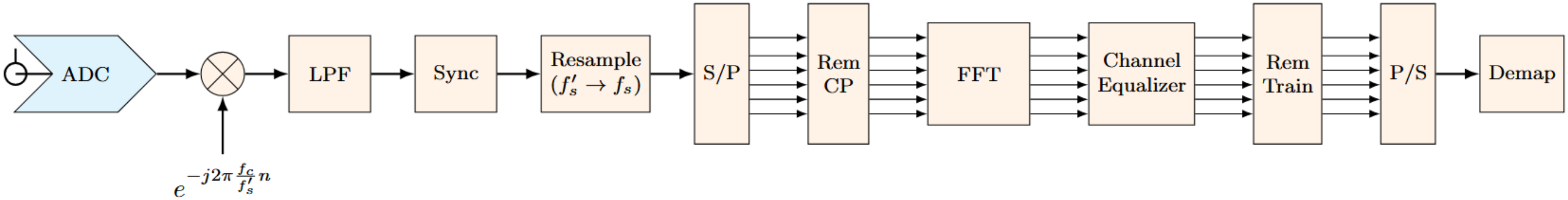
The transmitter

- Block diagram of the transmitter

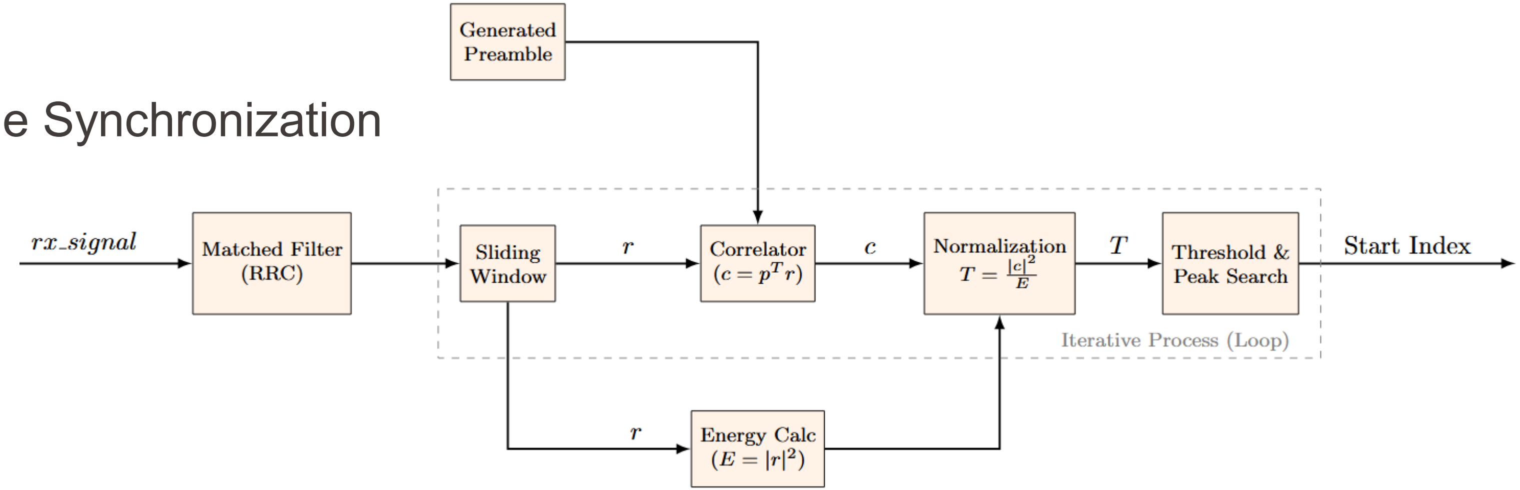


The receiver

- Block diagram of the receiver

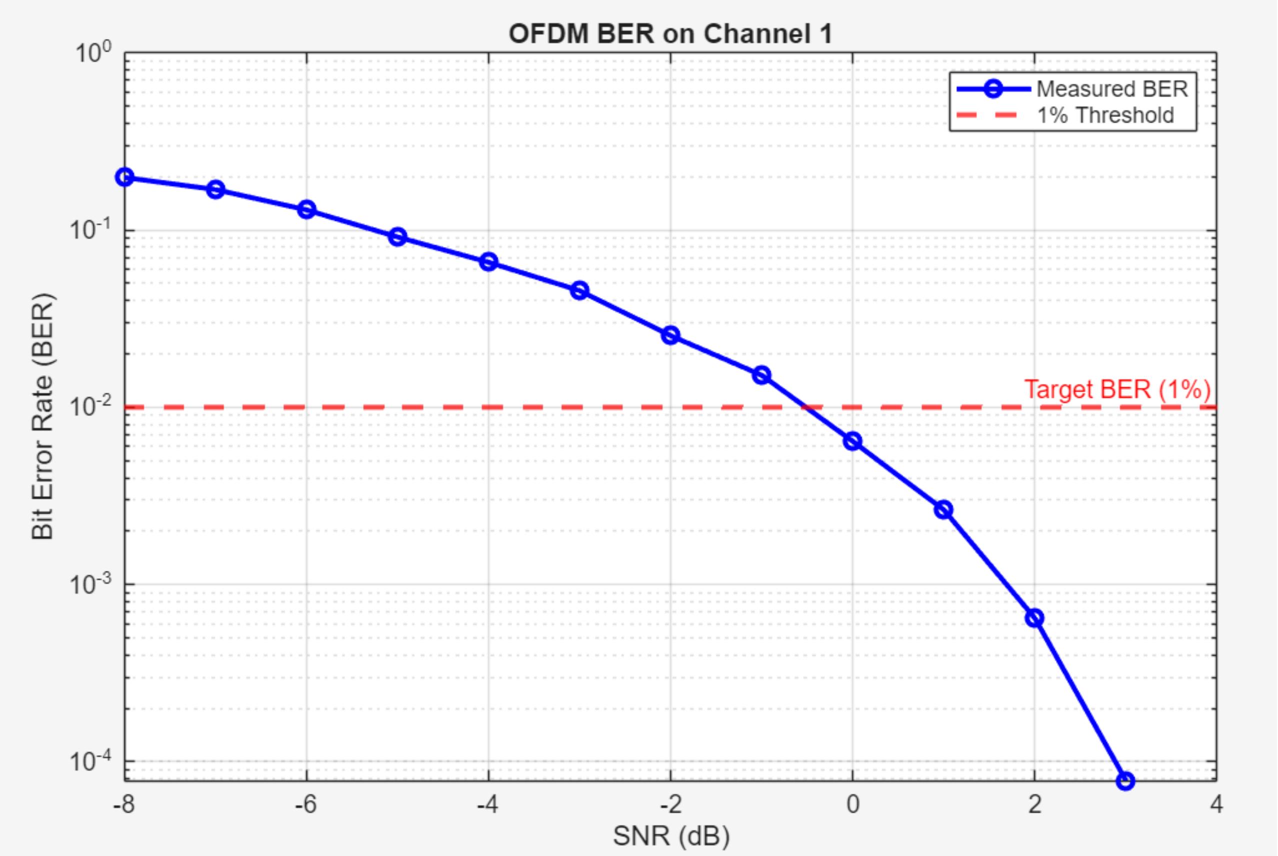


- Frame Synchronization



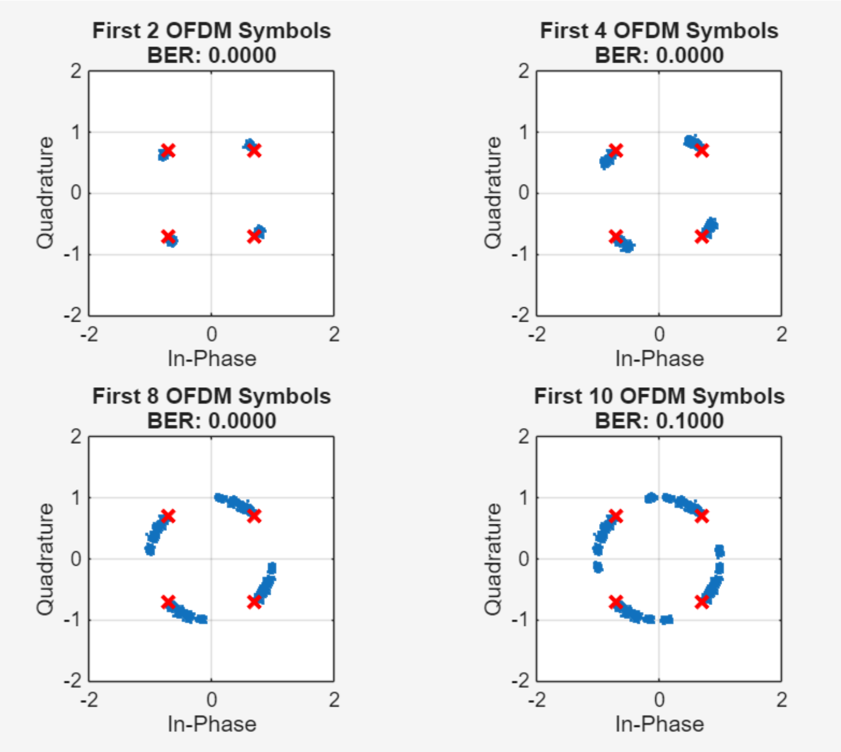
OFDM transmission on Channel 1

- Maximum SNR to have BER < 1%
- SNR = 0 dB BER < 1%
- SNR = -1 dB BER> 1%

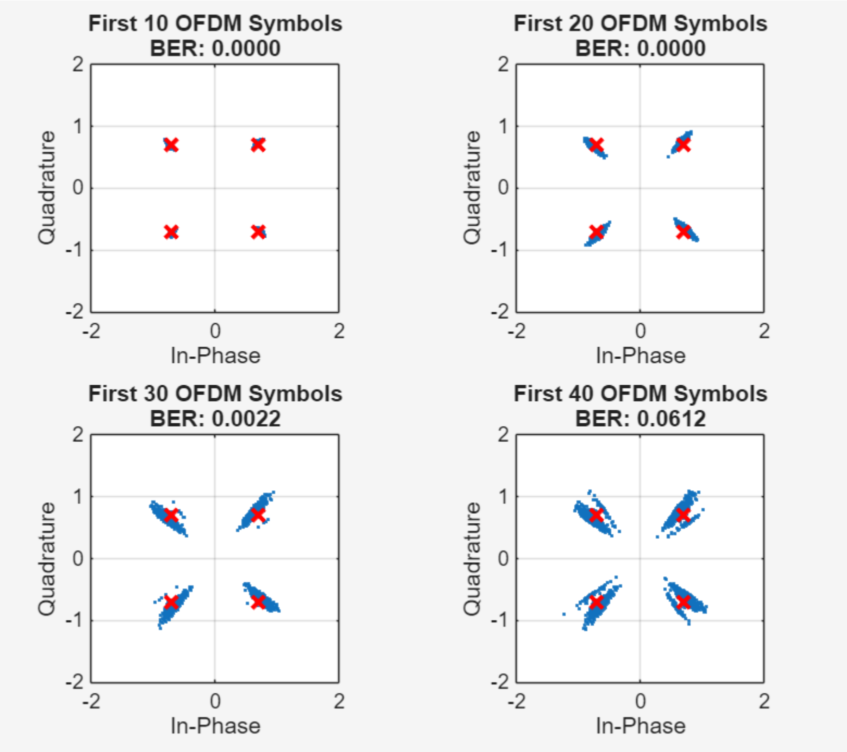


OFDM transmission on Channel 2

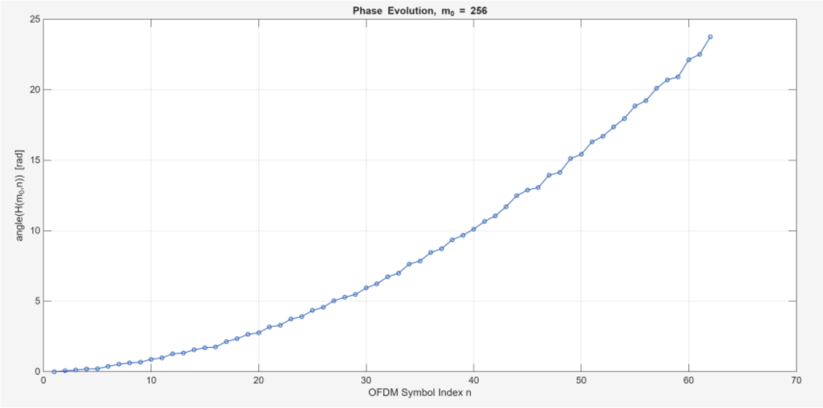
- Transmission without phase tracking

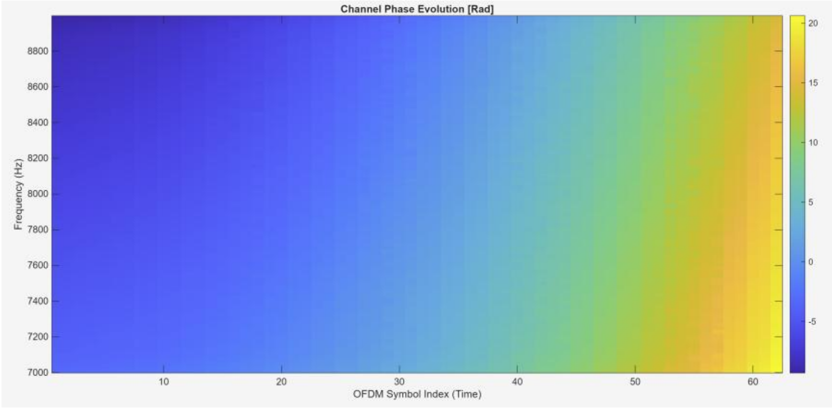


- Transmission using Viterbi-Viterbi

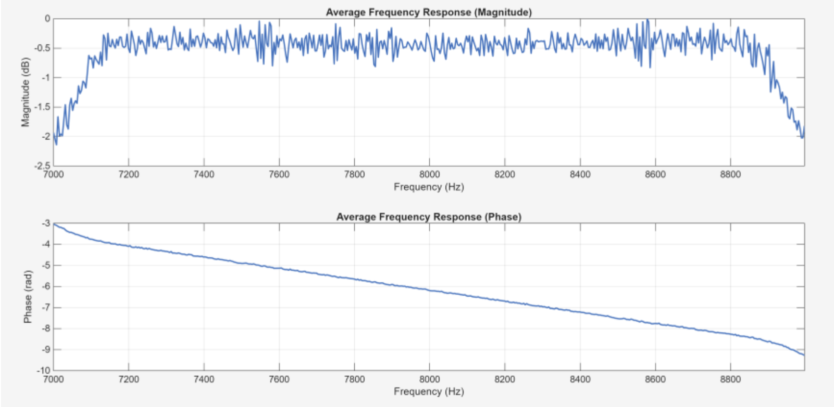


Channel 2 Characteristics

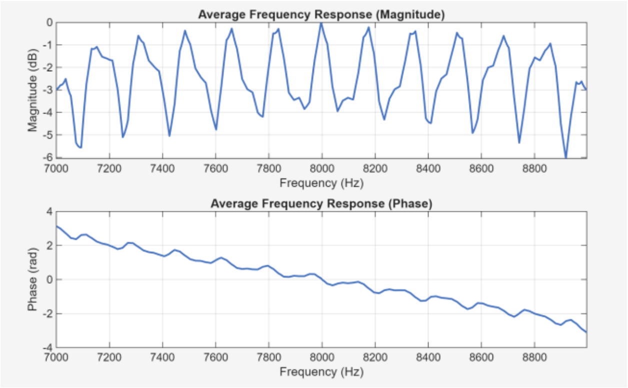


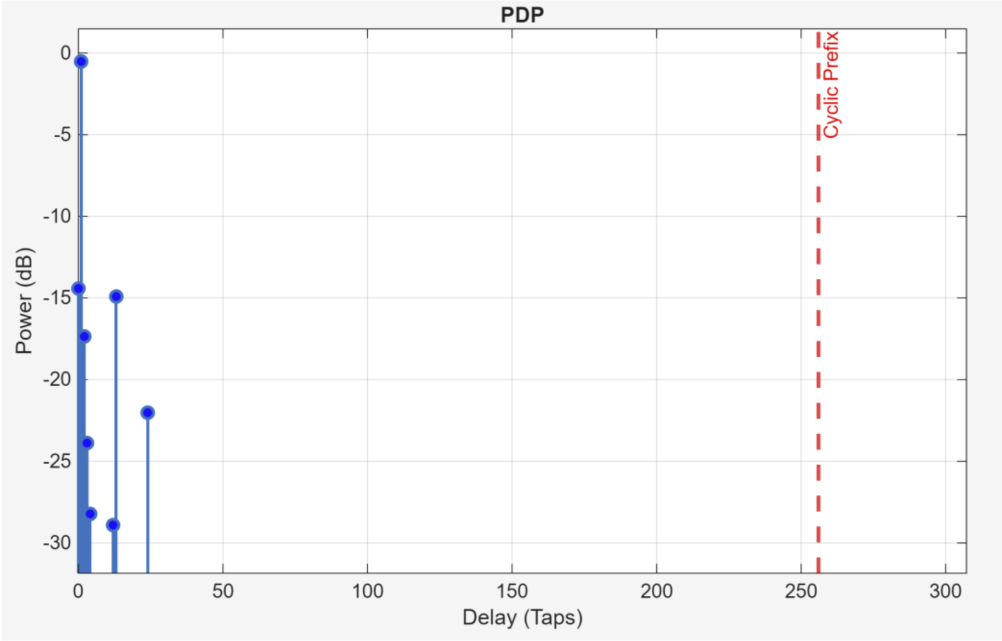


- Carrier frequency offset:
  - Linear phase drift over time



Channel 3 Characteristics

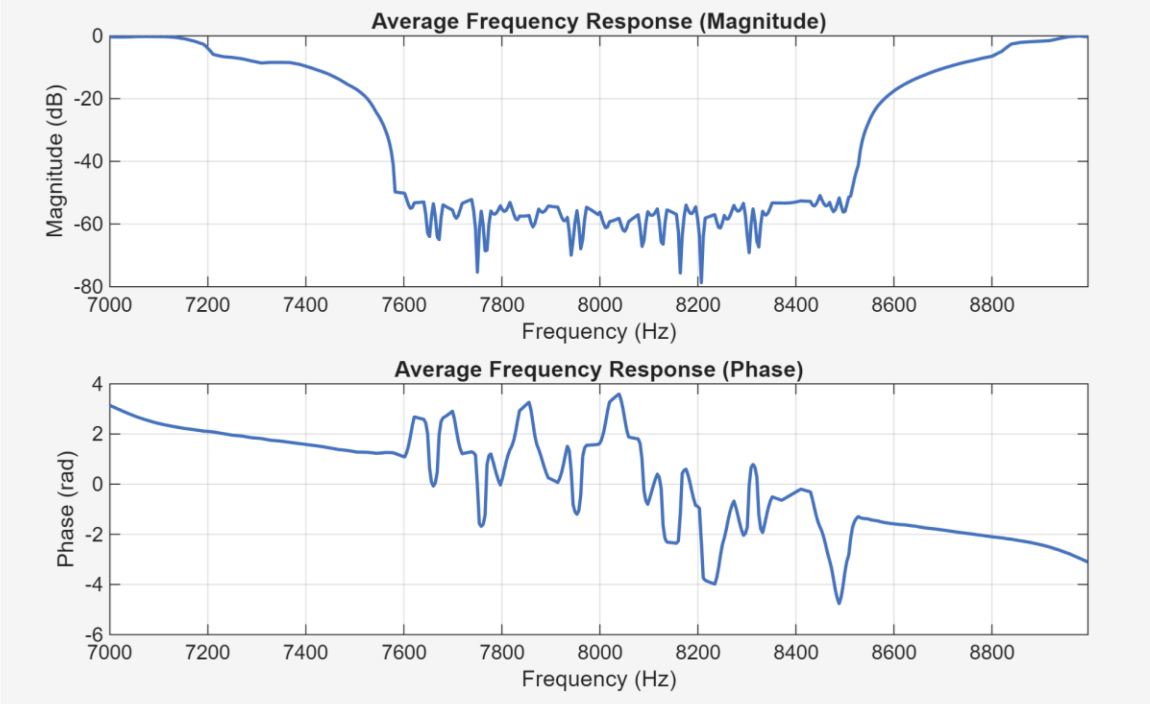




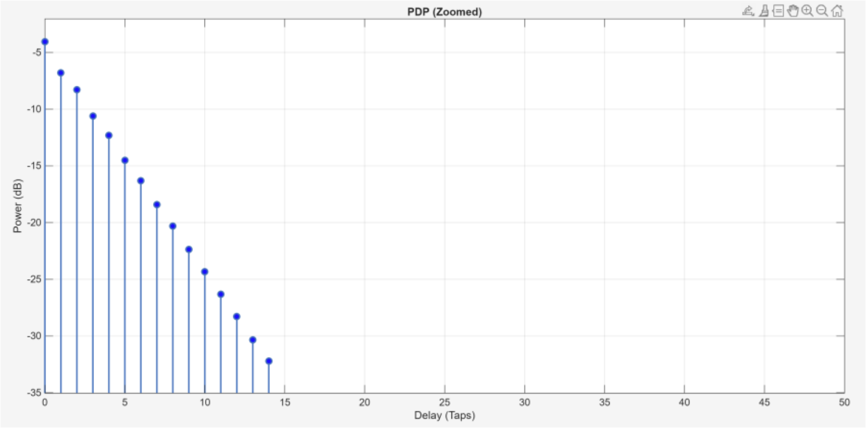
- Multipath channel:
  - Channel shorter than CP, so no ISI occurred
- Static amplitude and phase of the channel

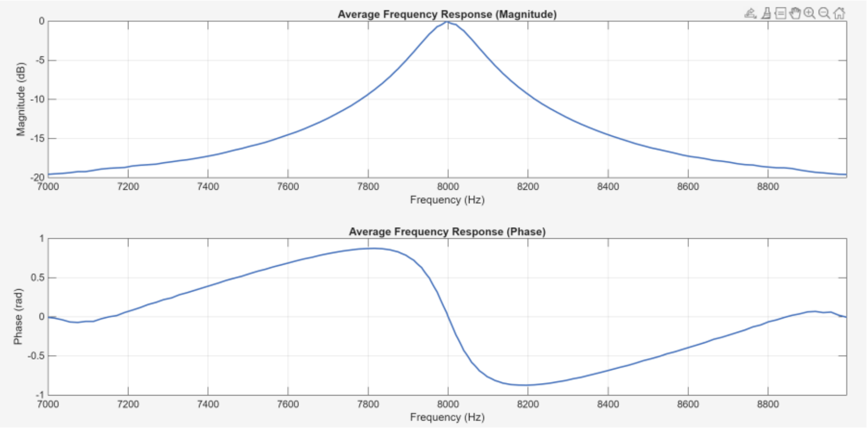
Channel 4 Characteristics

- Static band-stop filter:
  - BER of approximately 25%
  - Half of the subcarriers is destroyed

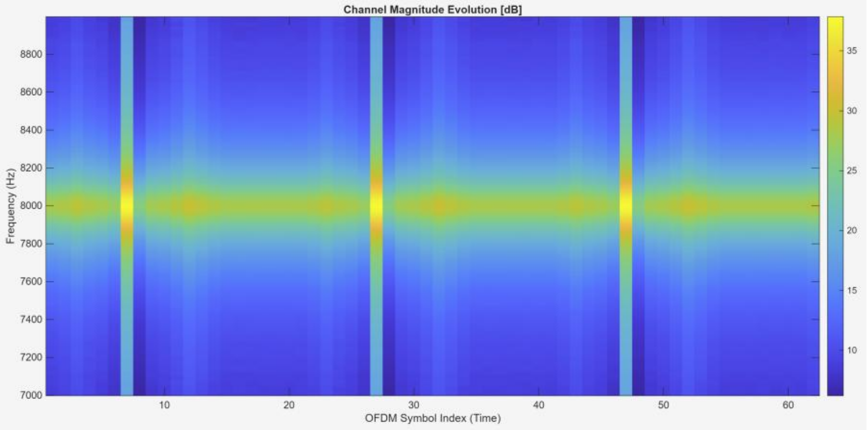


Channel 5 Characteristics





- Time-varying magnitude:
  - Regular impulses in channel magnitude
- Many significant taps in PDP:
  - Exponential decay.
  - Still much shorter than CP



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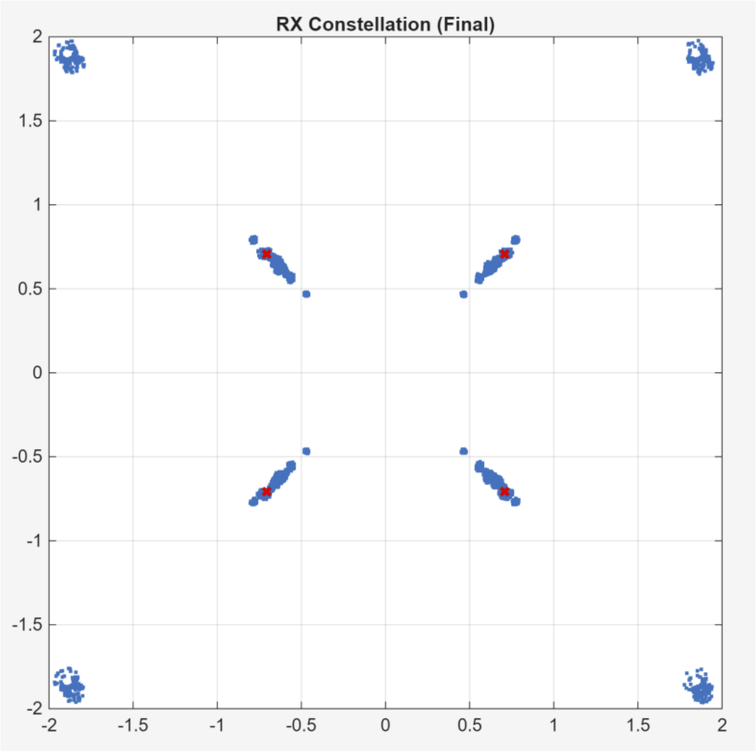
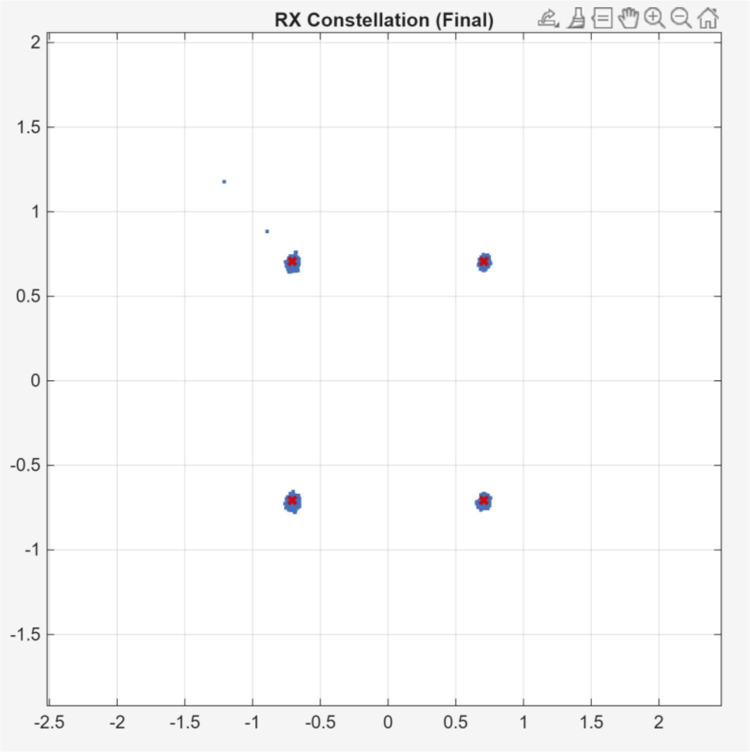
Channel 5 effects

- Transmission with Block estimation:



- Transmission with Comb estimation:



Spectral Efficiency

$$\eta = \frac{R_{net}}{B} \quad [\mathrm{bit/s/Hz}]$$

- Bandwidth (B) = 2000 Hz
- Net Data Rate (R_{net})
- Modulation QPSK (2 bits/symbol)
- Cyclic Prefix (N_{cp}) = 256

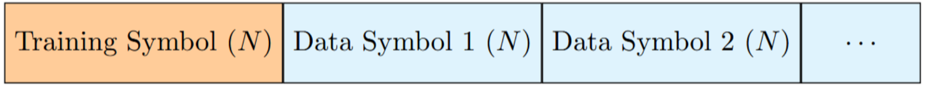
- Theoretical Max ≈ 1.33 bit/s/Hz (ignoring frame overhead)

To increase the Spectral Efficiency:

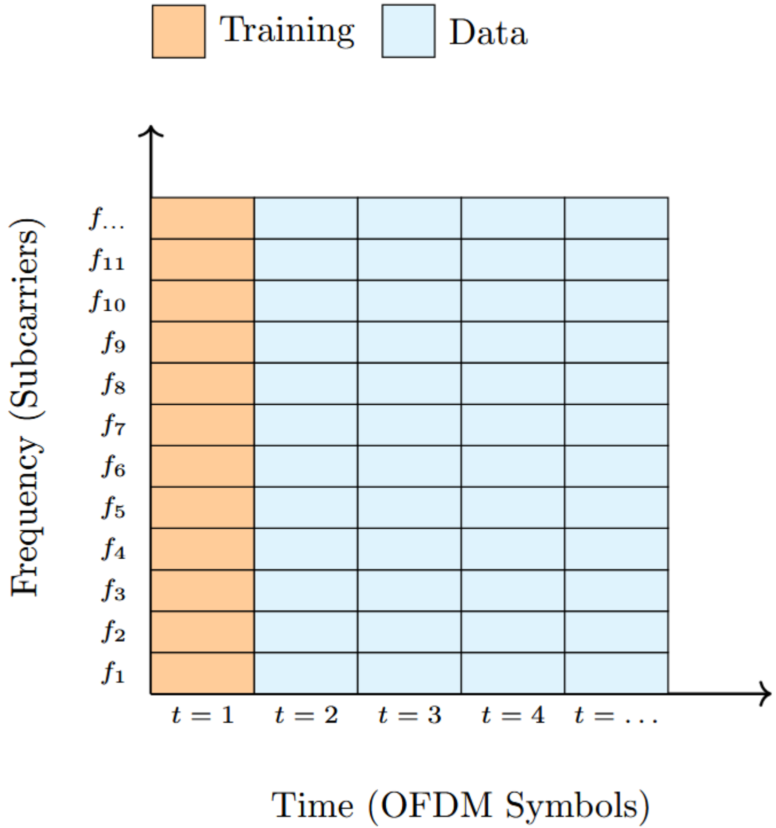
- Increase Modulation Order (16-QAM): Doubles bits per subcarrier, but needs more SNR
- Reduce Cyclic Prefix Length: Increases the risk of Inter-Symbol Interference

Channel Estimation: Block

- Training Symbol inserted before Data



- Estimate the channel once per block
- Works providing the channel doesn't change much
- Estimate $H = \frac{Y}{X}$
- We also added a version with Viterbi phase estimation

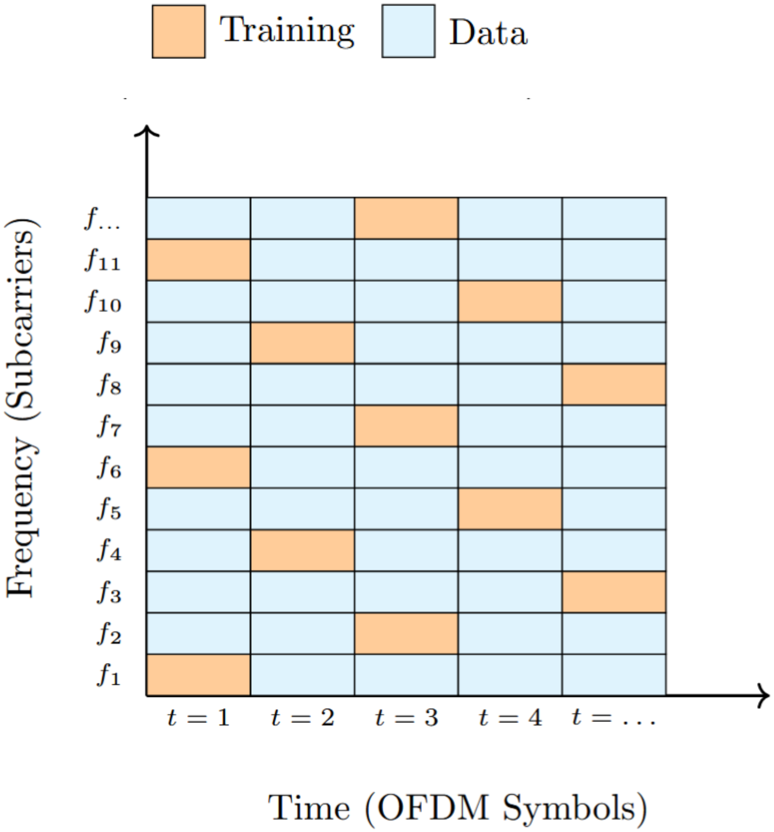


Channel Estimation: Comb

- 1 Pilot every 4 Data (L = 5) inserted serially

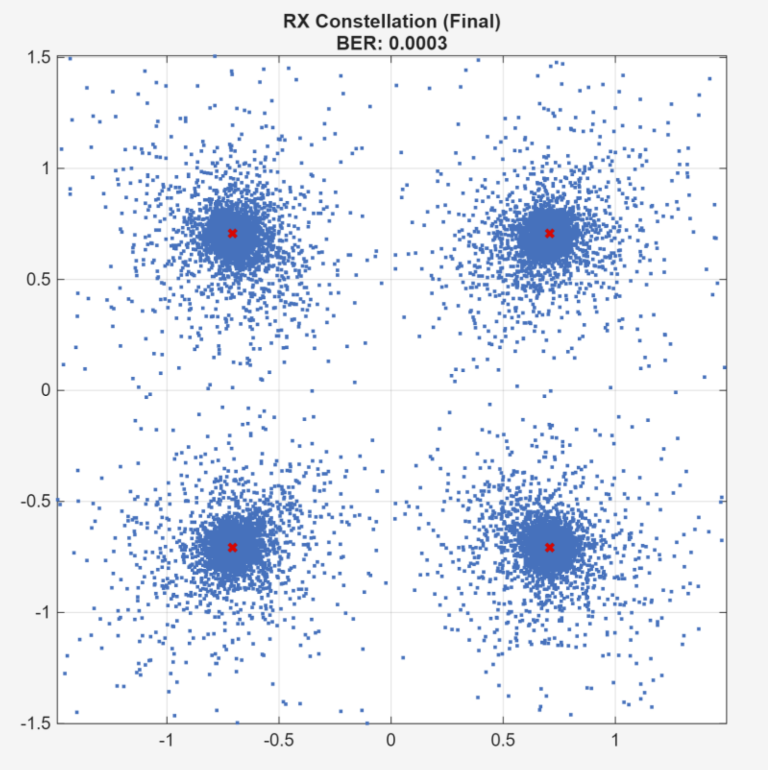
P_1	$D_1 \dots D_4$	P_2	$D_5 \dots D_8$	P_3	$\cdots$

- Estimate the channel continuously
- Extract pilot subcarriers at known indices
- Estimate $H = \dfrac{Y}{X}$
- Linear Interpolation to recover the channel for data subcarriers



Comb vs Block transmission on Channel 2

- Transmission using Comb estimation



- Transmission using Block + Viterbi

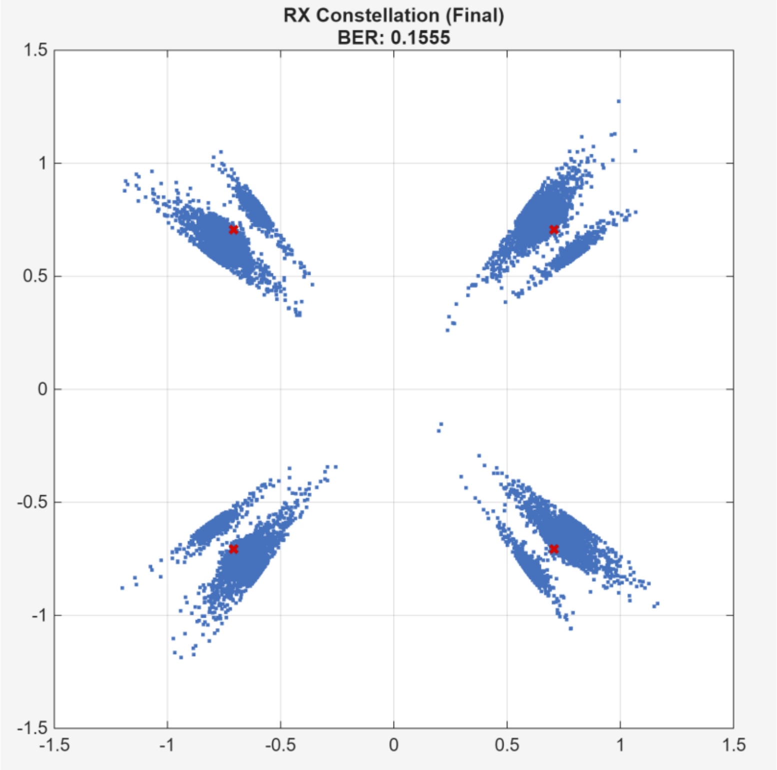
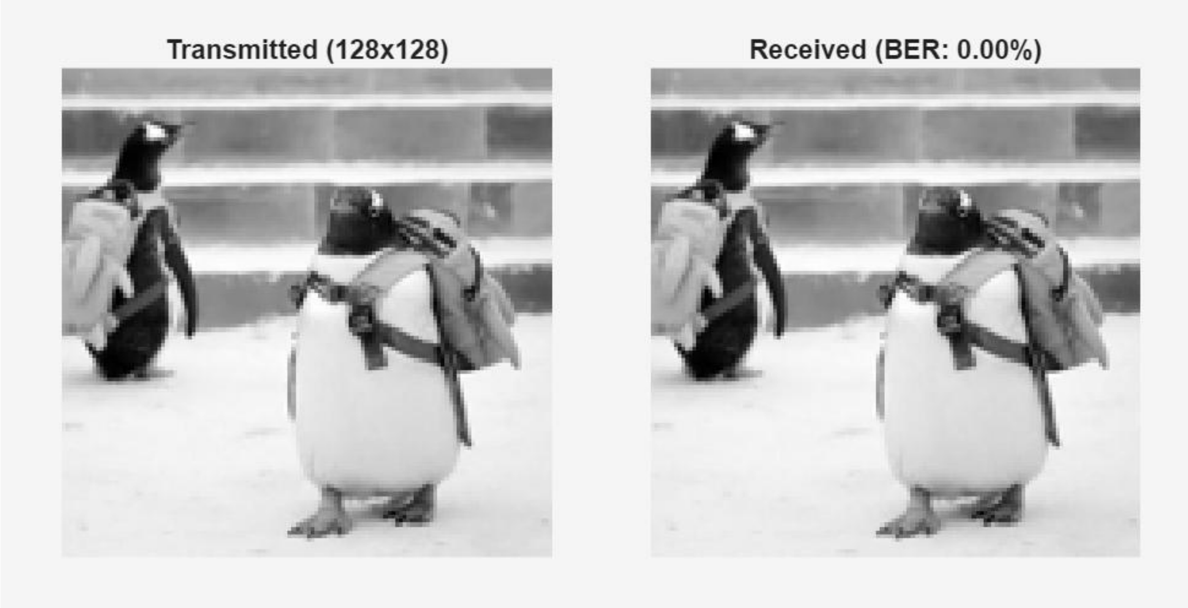


Image Transmission

- Resized to 128x128 pixels to reduce time transmission
- Converted to binary stream
- Method: Comb Pilot
- In random bitstreams, every error is equal
- In images an error in the MSB changes a pixel from black to white



Throughput Maximization

- R = Throughput (bits per second)
- N_{sc} = Number of Subcarriers
- B = Bandwidth
- M = Modulation Order
- N_{cp} = Cyclic Prefix

$$R = \frac{N_{sc} \cdot \log_2(M)}{T_{total}} = \frac{N_{sc} \cdot \log_2(M) \cdot B}{N_{sc} + N_{cp}}$$

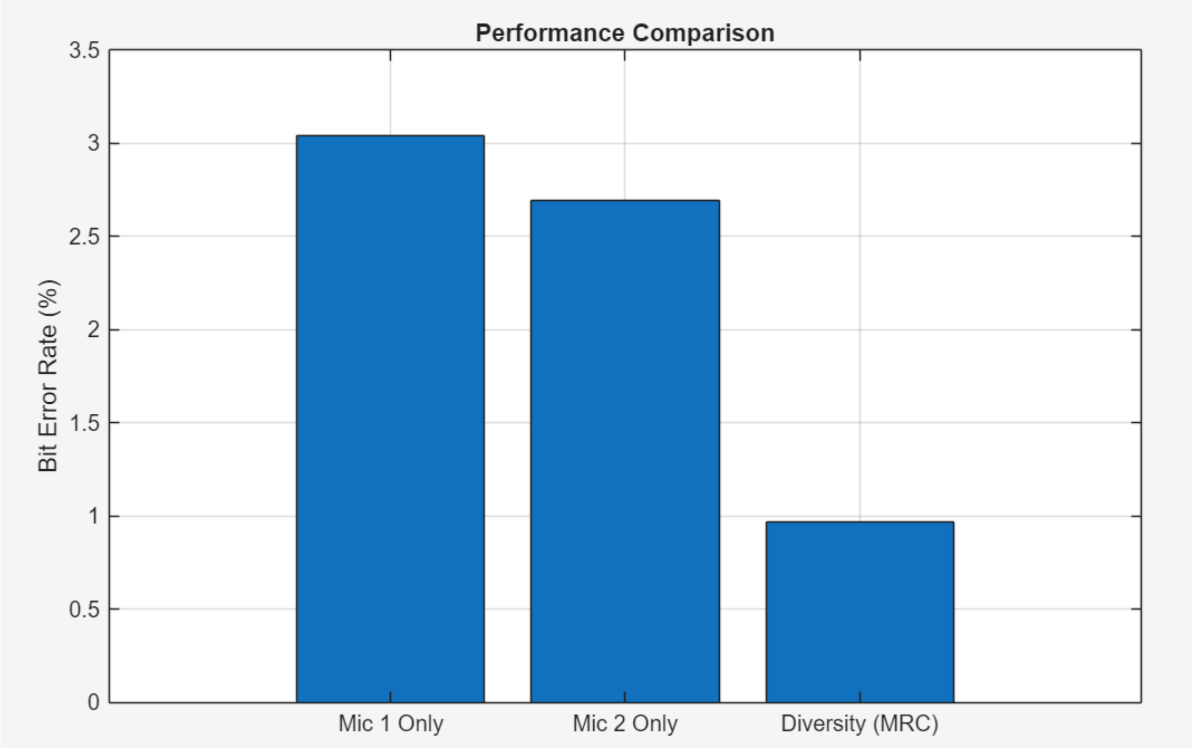
- Increased Bandwidth
- Reduce N_{cp} Cyclic Prefix
- Carrier frequency to 12 kHz

Bandwidth	Cyclic Prefix	Throughput	BER
16 kHz	128	25600 bps	6.76 %
16 kHz	32	30118 bps	9.95 %
16 kHz	0	32000 bps	13.78 %

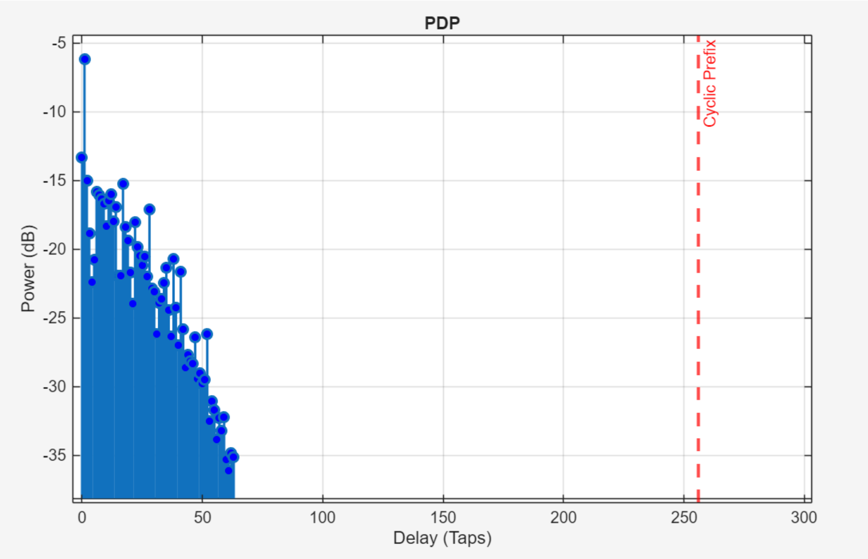
Receiver Diversity (MRC)

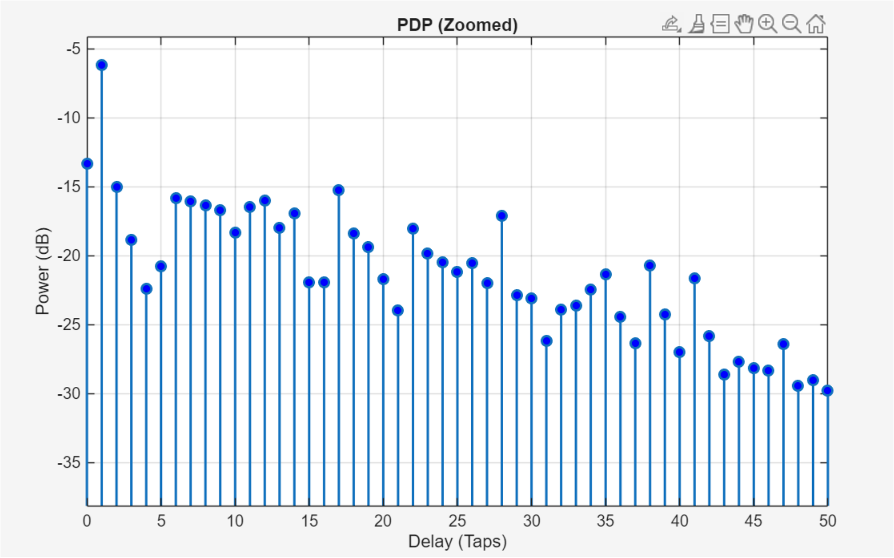
- Diversity improved BER with respect to single receivers
- Microphone of the computer (Mic 1)
- USB Microphone (Mic 2)

$$\hat{S} = \frac{H_1^* Y_1 + H_2^* Y_2}{|H_1|^2 + |H_2|^2}$$



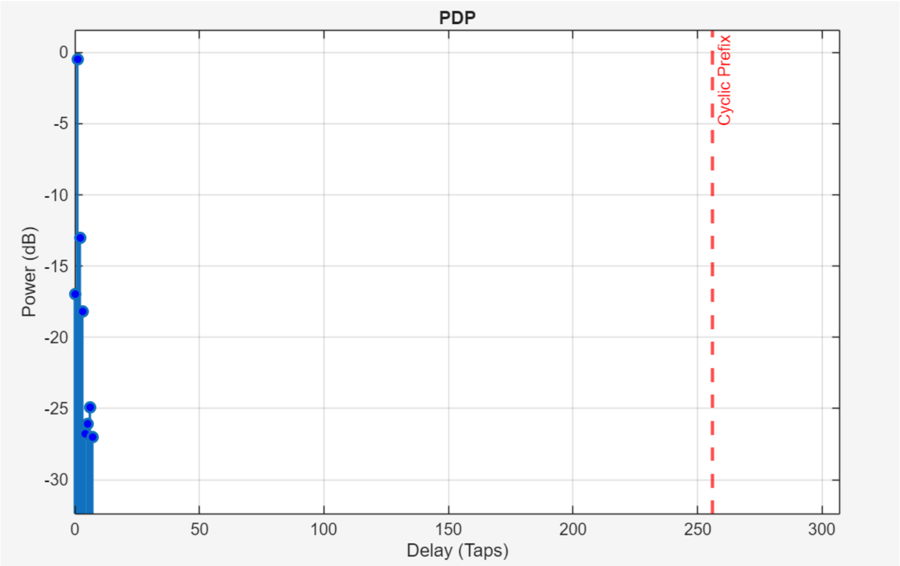
Transmission in CO building toilet

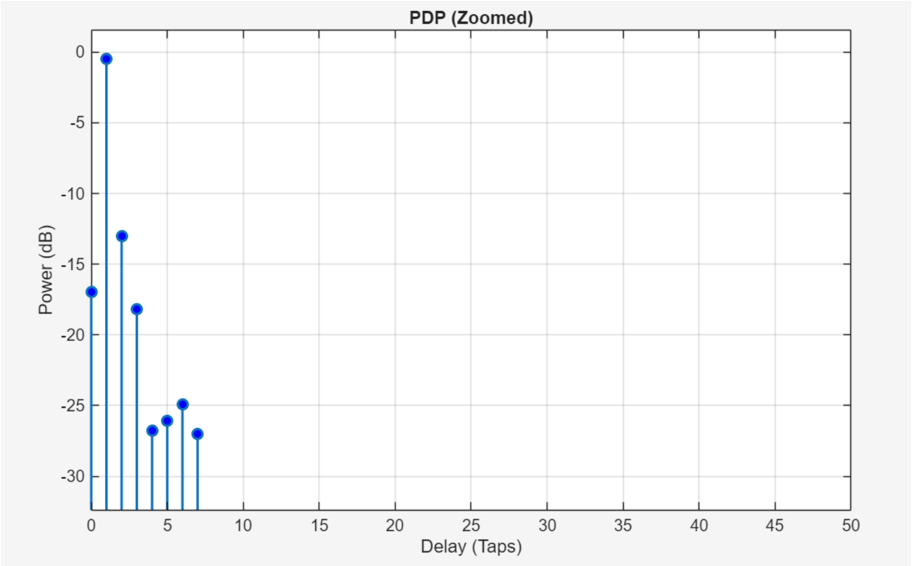




- Channel affected by noise and echo (toilet's narrow hallway)
- Longer PDP than previous estimated channels

Transmission outside CO building





- Outdoor the transmission was cleaner
- Short PDP
- BER = 0%

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Thank you

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